LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Business Intelligence and Data Summarisation using AI Prompts

Project Domain

Excel or Power BI with Chatbot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship aims to equip students with practical skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

The primary objective of this project is to analyze financial and economic data to understand the relationship between macroeconomic indicators and market performance. The dataset used in this project includes variables such as GDP growth, inflation rate, unemployment rate, interest rates, stock market index values, and commodity prices. The major problem identified was the presence of unstructured and inconsistent data, which made analysis difficult and time-consuming.

To address this issue, data cleaning and preprocessing techniques were applied using Microsoft Excel. This included removing duplicate entries, handling missing values, and standardizing data formats. After cleaning the dataset, data analysis was performed to identify trends and relationships between economic indicators and financial market behavior.

Microsoft Power BI was used to create interactive dashboards and visualizations, enabling clear and effective presentation of insights such as market trends, economic fluctuations, and key performance indicators. Additionally, AI tools such as ChatGPT and Microsoft Copilot were used to assist in data interpretation, summarization, and report generation.

The results of the project show that proper data cleaning and visualization significantly improve the accuracy and usability of financial data. The project successfully demonstrates how combining data analytics with AI tools can simplify complex datasets and support better decision-making.
In conclusion, this project highlights the importance of data-driven analysis in understanding economic and financial trends and showcases the practical application of modern analytical tools in real-world scenarios.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

The main problem addressed in this project is the difficulty in analysing large financial and economic datasets to understand the relationship between macroeconomic indicators and financial market performance*. Financial data such as stock index prices, GDP growth, inflation rate, unemployment rate, interest rates, and commodity prices are often available in large volumes, making it challenging to extract meaningful insights without proper data processing and visualization tools.*

This problem is significant because economic indicators directly influence financial markets, investment decisions, and business strategies*. If these indicators are not analysed properly, organizations and analysts may fail to identify important market trends, which can lead to poor financial decisions, inaccurate forecasts, and inefficient resource allocation. In the real world, investors, financial analysts, policymakers, and business organizations rely on accurate data analysis to understand economic conditions and predict future market behaviour.*

The groups most affected by this problem include financial analysts, investment firms, economic researchers, business organizations, and policymakers. These stakeholders need reliable insights from economic data to support strategic planning, investment decisions, and economic policy development.

However, existing approaches to analysing financial data often have several limitations. Many traditional methods rely heavily on manual data processing, which is time-consuming and prone to human error. Large datasets may contain inconsistencies, missing values, or formatting issues that make analysis difficult. Additionally, without proper visualization tools, it becomes challenging to clearly identify patterns, trends, and relationships within the data.

Therefore, this project focuses on addressing these challenges by applying data cleaning, analysis, and visualization techniques using tools such as Excel, Power BI, and AI-assisted technologies. By transforming raw financial data into clear visual insights, the project aims to make economic data easier to understand and more useful for decision-making.

1.2 Motivation

The motivation for choosing this project comes from my strong interest in the field of business and economics. I believe that business plays a very important role in our lives because it drives economic development, creates employment opportunities, and supports the growth of industries and organizations. In many ways, business can be compared to the lifeblood of society, as it influences almost every aspect of our daily life, from production and trade to financial stability and

economic growth. This belief inspired me to explore a project related to finance and economic data analysis.

From a personal and professional perspective, this project is meaningful because it helps me understand how business and economic activities are influenced by various financial indicators such as GDP growth, inflation rates, interest rates, unemployment levels, and stock market performance. By studying and analysing these factors, I can gain a deeper understanding of how the economy functions and how businesses respond to changing economic conditions. This knowledge is valuable for building a strong foundation in business and finance-related careers.

In the real world, the analysis of financial and economic data has many practical applications. Businesses, financial institutions, investors, and policymakers rely on economic data to make informed decisions, plan strategies, and predict future market trends. *By transforming raw data into meaningful insights through data analysis and visualization, organizations can better understand economic patterns, manage risks, and improve their decision-making processes.*

This project can also contribute to the financial and business domain by demonstrating how modern data analytics tools can simplify complex datasets and present them in an understandable format. Visual dashboards and analytical reports can help organizations identify trends and patterns that may not be easily visible through traditional methods.

Furthermore, this project aligns with modern industry trends where artificial intelligence and data analytics are increasingly used to support business decisions. Many organizations are adopting AI-powered tools and business intelligence platforms to analyse large datasets quickly and efficiently. By using data analytics techniques along with AI-assisted tools, this project reflects the growing importance of technology-driven analysis in the business and financial sectors.

1.3 Objectives

- *To* ***understand the structure and content of the financial and economic dataset*** *used in the project.*
- *To* ***clean and preprocess the dataset by removing duplicates, correcting formatting errors, and handling missing values.***
- *To* ***analyse the relationships between financial market indicators and macroeconomic factors such as GDP growth, inflation, unemployment, and interest rates.***
- *To create visualizations and dashboards that clearly present trends and patterns within the dataset.*
- *To* ***use AI tools such as Copilot or AI chatbots to assist in data analysis and report generation.***
- *To generate meaningful insights from the data that can help understand economic and financial trends.*

1.4 Scope of the Project

*The scope of this project focuses on analysing a finance and economics dataset to understand the relationship between different economic indicators and financial market performance**. The dataset includes variables such as GDP growth, inflation rate, unemployment rate, interest rates, stock market index values, and commodity prices. By analysing these indicators, the project aims to identify patterns and trends that help in understanding how economic factors influence financial markets and business activities.***

The project mainly involves data cleaning, data analysis, and data visualization using tools such as Microsoft Excel, Power BI, and AI-assisted platforms. *The raw dataset is first cleaned by removing duplicate entries, correcting data formats, and handling missing values. After preparing the dataset, analytical techniques and visualization tools are used to present the data in the form of charts, graphs, and dashboards that clearly highlight important insights.*

This project is limited to exploratory data analysis and visualization of the available dataset. ***It does not include advanced predictive models, machine learning algorithms, or real-time financial forecasting systems.*** *The tools used are limited to the ones assigned for the fellowship project, such as Excel, Power BI, and AI tools like Copilot or AI chatbots.*

The project is intended primarily for educational purposes as part of the AICW Fellowship Program. Its goal is to demonstrate how modern data analytics and AI-assisted tools can be used to simplify complex financial data and support better understanding of economic trends. Although the project may provide useful insights, it is not designed as a full-scale professional financial analysis system used by organizations.

CHAPTER 2

Literature Survey

A literature survey is an important part of any research project because it helps to understand the existing knowledge, research studies, and technologies related to the project domain. It provides an overview of how researchers and professionals have previously addressed similar problems and what methods or tools they used. In the field of finance and economics, large datasets are commonly analysed to study market behaviour, economic growth, financial stability, and business performance.

Financial and economic data analysis has gained significant importance in recent years because organizations rely heavily on data-driven decision making. Governments, financial institutions, investors, and businesses continuously analyse economic indicators such as Gross Domestic Product (GDP), inflation rate, unemployment rate, interest rates, stock prices, and commodity prices to evaluate economic conditions and make strategic decisions.

This chapter reviews five relevant research studies and compares their approaches with the methodology used in this project. The review also highlights the tools and technologies commonly used in financial data analysis and identifies gaps that this project attempts to address.

2.1 Existing Work and Research

Research Paper 1: Stock Returns, Expected Returns, and Real Activity

A well-known study conducted by Eugene F. Fama examines the relationship between stock market returns and real economic activities. The main objective of the research was to understand how economic growth and business cycle changes influence stock market performance. The study analyze various economic indicators such as industrial production growth, economic output, and financial market returns to determine whether real economic activities can explain stock price movements.

The research used statistical analysis and regression models to examine long-term relationships between stock returns and economic variables. The results indicated that stock market returns are closely connected with economic growth. When economic activity increases, companies generally perform better, which leads to higher stock prices and better market performance. Conversely, economic slowdowns can negatively affect stock market returns.

Although the study provided strong theoretical and statistical evidence about the relationship between economic activity and stock market returns, it mainly focused on econometric models and complex statistical analysis. It did not focus on the use of visualization tools or simplified methods for interpreting financial data.

In comparison, the present project focuses on analyzing financial and economic data using practical tools such as Excel and Power BI. These tools help in creating visual dashboards and

charts that make it easier to understand the relationship between economic indicators and market performance.

Research Paper 2: The Time-Series Relations Among Expected Return, Risk, and Book-to-Market

Another important study by Eugene F. Fama and Kenneth R. French investigates the relationship between expected returns, financial risk, and the book-to-market ratio of companies. The main purpose of this research was to identify how financial risk factors influence stock returns over time.

The researchers analyze long-term financial market data and applied time-series econometric techniques to study the behavior of stock returns. Their findings showed that companies with higher risk and higher book-to-market ratios tend to generate higher expected returns in the long run. The study also demonstrated that risk factors play an important role in determining stock market performance.

The methodology used in this research involved advanced financial modelling and statistical techniques to evaluate the relationship between risk and return. These models helped investors and financial analysts understand how different financial variables affect investment outcomes.

However, the research mainly focuses on theoretical financial models and complex statistical analysis, which may be difficult for beginners to interpret. The study does not provide interactive visualization tools or practical dashboards for data interpretation.

The current project addresses this limitation by presenting financial data using simple analysis methods and visual dashboards. By using Excel and Power BI, the project allows users to explore financial relationships through charts and interactive visualizations.

Research Paper 3: Macroeconomic Factors and Stock Returns

A research study conducted by Andreas Humpe and Peter Macmillan focuses on understanding how macroeconomic variables affect stock market returns. The main objective of this research was to identify whether factors such as inflation, interest rates, exchange rates, and industrial production have a long-term relationship with stock market indices.

The study examined historical economic data from different financial markets and applied econometric techniques to analyze the relationship between macroeconomic variables and stock market performance. The results showed that macroeconomic indicators significantly influence stock prices and overall market trends.

The researchers used statistical and econometric models, including cointegration analysis and regression models, to study the long-term relationships between economic indicators and financial markets. These methods helped identify patterns and correlations between macroeconomic factors and stock market returns.

While the study provides important insights into financial market behavior, it mainly relies on complex econometric analysis. The research does not focus on presenting financial data in a visual or interactive format that can help users easily understand the results.

In contrast, the present project emphasizes data visualization and dashboard development using tools such as Excel and Power BI. These tools make it easier to interpret financial data and understand the impact of macroeconomic indicators on market performance.

Research Paper 4: Predicting Stock Market Index Using Machine Learning Techniques

With the advancement of technology, researchers have started applying machine learning techniques to financial market prediction. A study conducted by Jigar Patel, Sahil Shah, and Priyank Thakkar focuses on predicting stock market index movements using machine learning algorithms.

The objective of this research was to develop predictive models that could analyze historical stock market data and forecast future market trends. The researchers used machine learning techniques such as Artificial Neural Networks and Support Vector Machines to analyze stock market patterns.

The study demonstrated that machine learning models can provide better predictions compared to traditional statistical methods when large datasets are used. By training models on historical financial data, the system could identify patterns and relationships that influence stock market fluctuations.

However, the research requires advanced programming knowledge and machine learning expertise, which may not be suitable for beginner-level projects or users without technical backgrounds.

The current project takes a simpler approach by focusing on financial data analysis and visualization using accessible tools such as Excel and Power BI. Instead of complex predictive modeling, the project focuses on helping users understand financial trends through visual dashboards and data exploration.

Research Paper 5: Interactive Dynamics for Visual Analysis

Another important research study conducted by Jeffrey Heer and Ben Shneiderman focuses on the importance of interactive visualization techniques for analyzing complex datasets. The objective of this research was to develop methods that allow users to explore large datasets through interactive visual interfaces.

The study highlights how visualization tools such as interactive charts, graphs, and dashboards can improve the process of data analysis and decision-making. Instead of relying solely on statistical reports, visual tools allow users to explore patterns, trends, and relationships in data more effectively.

The researchers proposed frameworks for creating interactive visual analytics systems that combine human intuition with computational analysis. These systems allow users to manipulate data visually and gain insights quickly.

Although the research provides strong theoretical support for data visualization, it mainly focuses on visualization frameworks rather than specific applications in financial data analysis.

The current project applies these visualization principles directly to financial and economic datasets. By using tools such as Excel and Power BI, the project creates interactive dashboards that help users analyze financial indicators and understand market trends more efficiently.

Title of the Research Paper	Author(s)	Year of Publication	Problem Addressed	Approach / Solution Taken	Method Used	Scope of the Study
Stock Returns, Expected Returns, and Real Activity	Eugene F. Fama	1990	Understanding how economic activities influence stock market returns and financial market performance.	The study analyses macroeconomic indicators such as production growth and economic activity to determine their impact on stock returns.	Statistical analysis and regression modelling using economic and financial data.	Helps investors and economists understand the relationship between real economic activity and stock market behaviour.
The Time-Series Relations Among Expected Return, Risk, and Book-to-Market	Eugene F. Fama and Kenneth R. French	1999	Investigating how risk factors and financial ratios affect stock market returns over time.	The authors studied long-term stock market data to evaluate the relationship between expected return, risk, and book-to-market ratio.	Time-series econometric analysis and financial modelling.	Useful for portfolio management, risk analysis, and financial market research.
Macroeconomic Factors and Stock Returns	Andreas Humpe and Peter Macmillan	2009	Identifying the effect of macroeconomic indicators on stock market performance.	The study evaluates variables like inflation, industrial production, interest rates, and exchange rates to analyse stock returns.	Econometric analysis and cointegration modelling.	Helps policymakers, financial analysts, and researchers understand economic influences on stock markets.
Predicting Stock Market Index Using Machine Learning Techniques	Jigar Patel, Sahil Shah, and Priyank Thakkar	2015	Difficulty in predicting stock market trends using traditional statistical models.	The research applies machine learning techniques to forecast stock market movements using	Machine learning algorithms such as Artificial Neural Networks and	Provides advanced forecasting methods for investors and financial institutions.

				historical financial data.	Support Vector Machines.	
Interactive Dynamics for Visual Analysis	Jeffrey Heer and Ben Shneiderman	2012	Challenges in interpreting complex datasets without proper visualization techniques.	The research proposes interactive data visualization systems to analyse large datasets more effectively.	Interactive data visualization and dashboard-based analytics.	Useful in business intelligence, economic analysis, and decision-making environments.

Table:1 Exsisting Work and research.

2.2 Tools and Technologies Review

The reviewed research papers use various analytical tools and technologies to study the relationship between macroeconomic indicators and stock market performance. These tools mainly include statistical analysis methods, econometric modelling techniques, machine learning algorithms, and data visualization technologies. Each research paper focuses on a different approach to analyse financial and economic datasets.

For example, the study conducted by Eugene F. Fama mainly used statistical techniques and regression models to examine the relationship between economic activity and stock market returns. Similarly, the research work by Eugene F. Fama and Kenneth R. French applied time-series analysis and financial modelling techniques to study how risk factors influence expected returns in financial markets.

Another study by Andreas Humpe and Peter Macmillan used econometric tools such as cointegration and regression analysis to analyse the impact of macroeconomic variables on stock market indices. In contrast, the research conducted by Jigar Patel and his co-authors introduced machine learning technologies like Artificial Neural Networks and Support Vector Machines to predict stock market trends using historical financial data.

Furthermore, the study by Jeffrey Heer and Ben Shneiderman focused on interactive data visualization technologies that help users analyse large datasets through dashboards, charts, and graphical interfaces.

Overall, these studies demonstrate that different analytical tools and technologies are used to analyse financial data depending on the objectives of the research. The following table presents a comparative summary of the tools and technologies used in each of the reviewed research papers.

Here is a ***tabular column showing the tools and technologies used in the reviewed research papers****.*

Research Paper Title	Author(s)	Year	Tools / Technologies Used	Purpose of the Tools
Stock Returns, Expected Returns, and Real Activity	Eugene F. Fama	1990	Statistical Analysis, Regression Models, Time-Series Analysis	To analyse the relationship between economic activity and stock market returns.
The Time-Series Relations Among Expected Return, Risk, and Book-to-Market	Eugene F. Fama and Kenneth R. French	1999	Econometric Modelling, Financial Modelling, Time-Series Analysis	To study how risk factors and financial ratios influence expected stock returns.
Macroeconomic Factors and Stock Returns	Andreas Humpe and Peter Macmillan	2009	Cointegration Analysis, Regression Analysis, Econometric Models	To examine the relationship between macroeconomic indicators and stock market indices.
Predicting Stock Market Index Using Machine Learning Techniques	Jigar Patel, Sahil Shah, and Priyank Thakkar	2015	Artificial Neural Networks (ANN), Support Vector Machines (SVM), Machine Learning Algorithms	To predict stock market trends using historical financial data.
Interactive Dynamics for Visual Analysis	Jeffrey Heer and Ben Shneiderman	2012	Data Visualization Tools, Interactive Dashboards, Visual Analytics	To analyse large datasets using interactive charts and visualization techniques.

Table:2 Tools and Technologies used by the researchers.

2.3 Gaps Identified

After reviewing the selected research papers, several important research gaps were identified. These gaps highlight the limitations of previous studies and justify the need for the current project.

1. Lack of User-Friendly Tools
The research by Eugene F. Fama mainly focuses on statistical and econometric analysis to study the relationship between economic activity and stock market returns. However, the study does not provide simple tools or visual dashboards that allow users to easily interpret financial data.

2. Complex Financial Models
The study conducted by Eugene F. Fama and Kenneth R. French uses advanced financial models and time-series analysis to evaluate risk and expected returns. These complex techniques may be difficult for beginners and non-technical users to understand.

3. Limited Data Visualization
The research by Andreas Humpe and Peter Macmillan analyses macroeconomic factors affecting stock market returns using econometric methods. However, the study lacks interactive visualization tools that help present data insights clearly.

4. High Technical Requirements
The work of Jigar Patel, Sahil Shah, and Priyank Thakkar applies machine learning techniques such as neural networks and support vector machines. These approaches require programming knowledge and technical expertise, which may limit accessibility for general users.

5. Limited Application to Financial Dashboards

The research conducted by Jeffrey Heer and Ben Shneiderman focuses on interactive visualization frameworks, but it does not specifically apply these visualization methods to financial datasets and economic indicators.

Figure:1 Research gap identified

CHAPTER 3

Proposed Methodology

3.1 System Design

The system is designed to convert raw financial and economic data into meaningful insights through a structured process involving multiple stages.

Data Source / Input

The dataset used in this project is a finance and business analytics dataset containing various economic and financial indicators such as GDP growth, inflation rate, unemployment rate, interest rates, stock index values, and commodity prices.

*The dataset is available in Excel format and consists of approximately **3000 rows and 24 columns**, where each row represents a data record and each column represents a specific variable or economic indicator.*

This structured dataset serves as the primary input for data cleaning, analysis, and visualization in the project.

Data Cleaning:

The raw dataset may contain inconsistencies such as missing values, duplicate records, and incorrect data formats. Data cleaning is performed using Microsoft Excel by:

- *Removing duplicate entries*
- *Handling missing values using appropriate techniques*
- *Standardizing data formats (dates, numerical values)*
- *Validating data for accuracy*

Analysis / Processing:

After cleaning, the dataset is analyze using:

- *Excel formulas and functions*
- *Pivot tables for summarization*
- *Power BI for advanced data modelling*
- *AI tools (Copilot/Chatbot) for generating insights and summaries*

The analysis focuses on identifying relationships between economic indicators and financial performance.

Visualization / Output:

The processed data is visualized using:

- *Charts (bar charts, line graphs, pie charts)*
- *Interactive dashboards in Power BI*
- *AI-generated summaries and insights*

These outputs help in clearly understanding trends and patterns in the dataset.

3.2 Modules / Components Used

The project is divided into the following key modules:

Module 1 — Data Ingestion:
Loading the dataset into Microsoft Excel and Power BI for further processing.

Module 2 — Data Cleaning:
Identifying and correcting errors such as duplicates, null values, and inconsistent formats.

Module 3 — Data Analysis:
Applying calculations, pivot tables, and AI-assisted analysis to derive insights.

Module 4 — Visualization / Report Generation:
Creating charts and dashboards to present the findings in a clear and interactive manner.

Module 5 — AI Integration:
Using AI tools like Copilot or ChatGPT to:

- *Generate summaries*
- *Suggest insights*
- *Assist in report writing*

3.3 Data Flow Diagram (DFD)

DFD Level 0 — Context Diagram:

Raw Dataset (Excel File)
↓
[Project System]
↓
Cleaned Data + Dashboard + Insights

DFD Level 1 — Detailed Process Flow:

Step 1: Data Input
↓
Step 2: Data Cleaning
(Remove duplicates, fix errors)
↓
Step 3: Data Analysis
(Pivot tables, formulas, AI tools)
↓
Step 4: Visualization
(Charts, Power BI dashboards)
↓
Step 5: Output
(Reports, Insights, Conclusions)

3.4 Advantages

The proposed methodology offers several advantages:

- *Reduces manual effort through automated data cleaning techniques*
- *Improves accuracy by eliminating errors and inconsistencies*
- *Provides clear and interactive visualizations for better understanding*
- *Uses AI tools to enhance analysis and save time*
- *Scalable approach that can be applied to other datasets and domains*
- *Cost-effective as it uses widely available tools like Excel and Power BI*

3.5 Requirement Specification

Hardware Requirements

Component	***Specification***
Processor	*Intel Core i3 or higher (i5 recommended)*
RAM	*Minimum 4 GB (8 GB recommended)*
Storage	*Minimum 50 GB free space*
Display	*1366 × 768 resolution or higher*
Internet	*Required for AI tools*

Table:3 Hardware Requirements

Software Requirements

Software	***Details***
Operating System	*Windows 10/11 or macOS*
Microsoft Excel	*Excel 2019 / Microsoft 365*
Power BI Desktop	*Latest version*
AI Tools	*ChatGPT / Microsoft Copilot*
Web Browser	*Google Chrome / Microsoft Edge*

Table:4 Software Requirements

CHAPTER 4

Implementation and Results

This chapter describes the practical implementation of the project and presents the results obtained after performing data cleaning, analysis, visualization, and AI-assisted processing. The aim of this chapter is to demonstrate how raw financial and economic data is transformed into meaningful insights using analytical tools.

4.1 Data Cleaning Results

The dataset used in this project was analyze and cleaned to ensure accuracy, consistency, and reliability for further analysis and visualization.

Dataset source:
*The dataset used in this project is a **Finance and Economics dataset**, provided as part of the project resources. It contains detailed financial, economic, and market-related indicators such as stock prices, GDP growth, inflation, and consumer spending.*

Original dataset size:
*The raw dataset consisted of **3000 rows and 24 columns**.*

Issues found:
After exploring the dataset, the following observations were made:

- *No **duplicate records** were found in the dataset*
- *No **missing (null) values** were present in any of the columns*
- *Most columns had **correct data types**, but verification was required*
- *Minor **formatting consistency checks** were needed (especially for date format and column naming clarity)*
- *Large numeric values required validation to ensure correctness and usability*

Cleaning steps applied:
The following steps were performed to prepare the dataset:

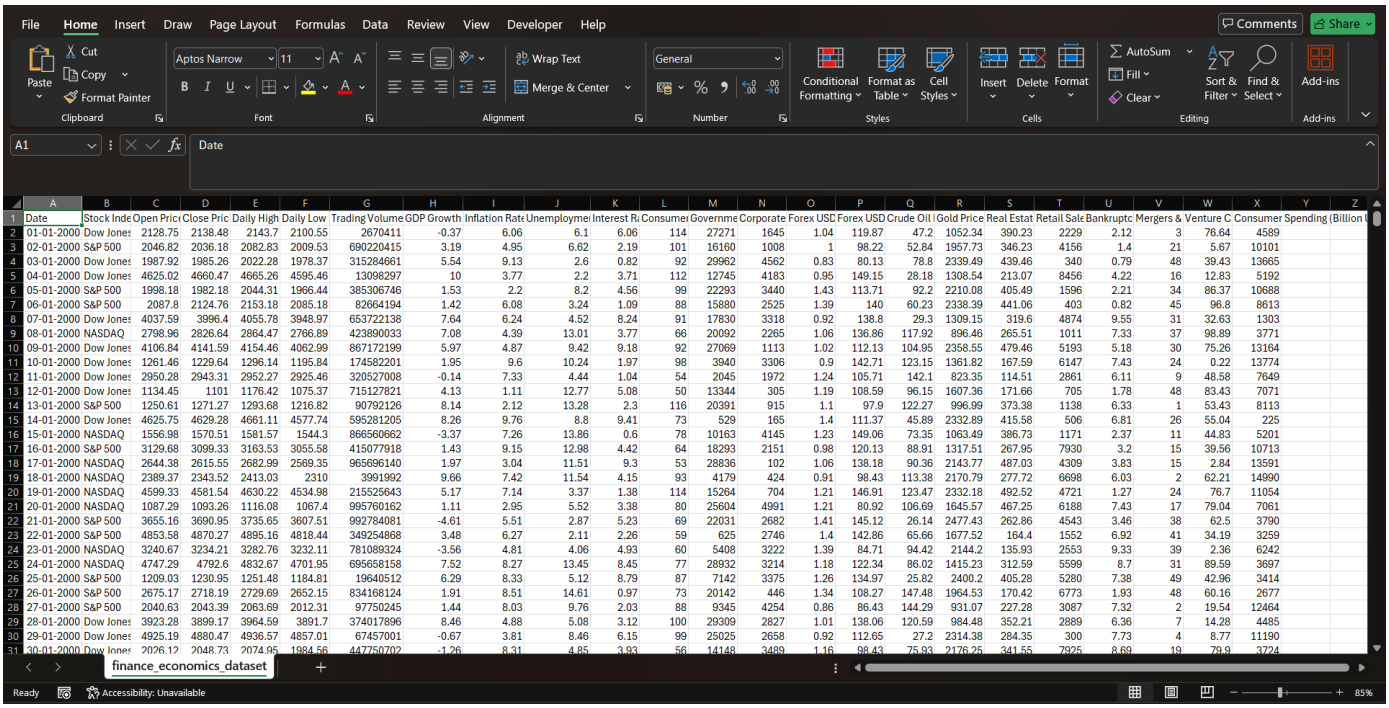
1. ***Checked for duplicates** using Excel tools – no duplicates found*
2. ***Validated missing values** using filters and functions – no null values detected*
3. ***Verified and corrected data types**, ensuring:*
    - *Date column was in proper **date format***
    - *Numeric columns were correctly set as **numbers/decimal values***
4. ***Standardized formatting** across columns for consistency*

5. ***Renamed or reviewed column headers*** *for better readability and understanding*

6. ***Validated numerical ranges*** *to ensure there were no unrealistic or extreme errors*

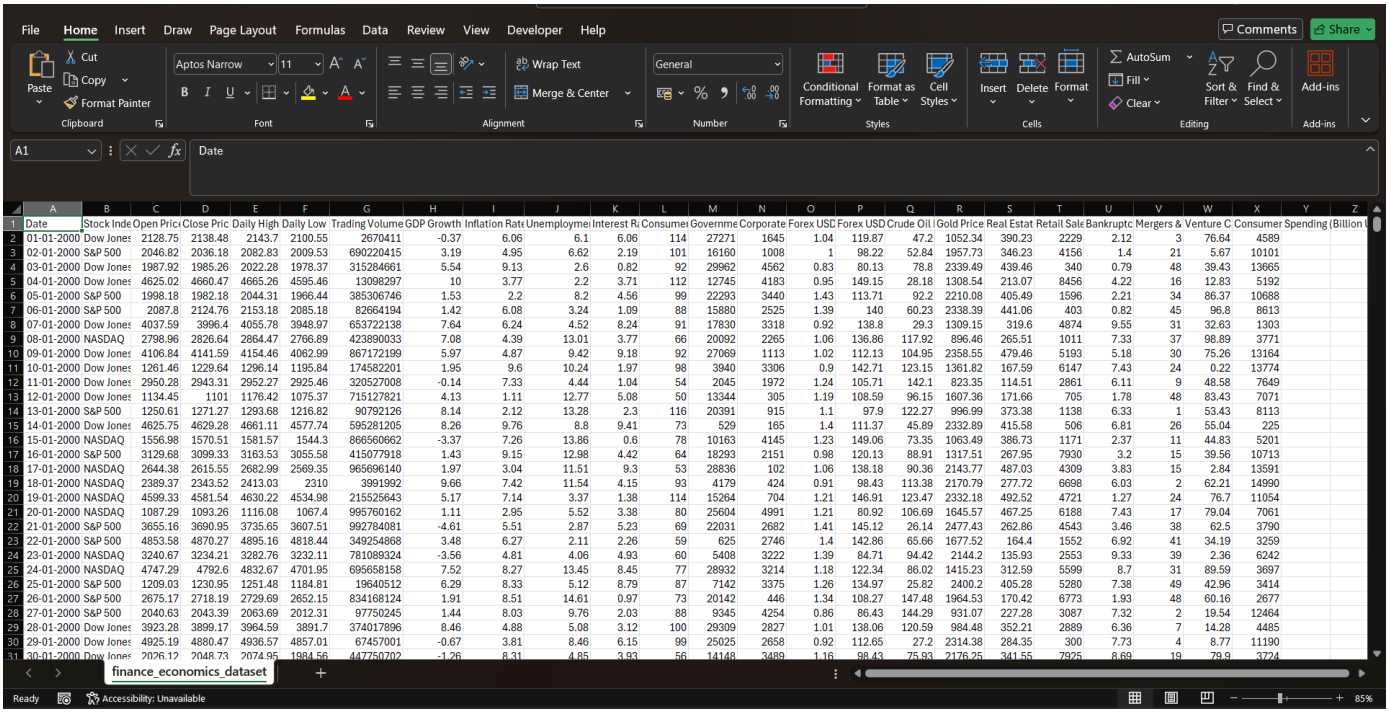
7. *Used* ***AI Chatbot assistance*** *to review dataset structure and confirm cleaning quality*

Cleaned dataset size:
Since no major inconsistencies were found, the dataset remained ***3000 rows and 24 columns*** *after cleaning.*



Date	Stock Inde	Open Pric	Close Pric	Daily High	Daily Low	Trading Volume	GDP Growth	Inflation Rate	Unemployme	Interest R	Consumer	Governme	Corporate	Forex USD	Forex USD	Crude Oil I	Gold Price	Real Estat	Retail Sale	Bankruptc	Mergers &	Venture C	Consumer	Spending (Billion U
01-01-2000	Dow Jones	2128.75	2138.48	2143.7	2100.55	2670411	-0.37	6.06	6.1	6.06	114	27271	1645	1.04	119.87	47.2	1052.34	390.23	2229	2.12	3	76.64	4589	
02-01-2000	S&P 500	2046.82	2036.18	2082.83	2009.53	690220415	3.19	4.95	6.62	2.19	101	16160	1008	1	98.22	52.84	1957.73	346.23	4156	1.4	21	5.67	10101	
03-01-2000	Dow Jones	1987.92	1985.26	2022.28	1978.37	315284661	5.54	9.13	2.6	0.82	92	29962	4562	0.83	80.13	78.8	2339.49	439.46	340	0.79	48	39.43	13665	
04-01-2000	Dow Jones	4625.02	4660.47	4665.26	4595.46	13098297	10	3.77	2.2	3.71	112	12745	4183	0.95	149.15	28.18	1308.54	213.07	8456	4.22	16	12.83	5192	
05-01-2000	S&P 500	1998.18	1982.18	2044.31	1966.44	385306746	1.53	2.2	8.2	4.56	99	22293	3440	1.43	113.71	92.2	2210.08	405.49	1596	2.21	34	86.37	10688	
06-01-2000	S&P 500	2087.8	2124.76	2153.18	2085.18	82664194	1.42	6.08	3.24	1.09	88	15880	2525	1.39	140	60.23	2338.39	441.06	403	0.82	45	96.8	8613	
07-01-2000	Dow Jones	4037.59	3996.4	4055.78	3948.97	653722138	7.64	6.24	4.52	8.24	91	17830	3318	0.92	138.8	29.3	1309.15	319.6	4874	9.55	31	32.63	1303	
08-01-2000	NASDAQ	2798.96	2826.64	2864.47	2766.89	423890033	7.08	4.39	13.01	3.77	66	20092	2265	1.06	136.86	117.92	896.46	265.51	1011	7.33	37	98.89	3771	
09-01-2000	Dow Jones	4106.84	4141.59	4154.46	4062.99	867172199	5.97	4.87	9.42	9.18	92	27069	1113	1.02	112.13	104.95	2358.55	479.46	5193	5.18	30	75.26	13164	
10-01-2000	Dow Jones	1261.46	1229.64	1296.14	1195.84	174582201	1.95	9.6	10.24	1.97	98	3940	3306	0.9	142.71	123.15	1361.82	167.59	6147	7.43	24	0.22	13774	
11-01-2000	Dow Jones	2950.28	2943.31	2952.27	2925.46	320527008	-0.14	7.33	4.44	1.04	54	2045	1972	1.24	105.71	142.1	823.35	114.51	2861	6.11	9	48.58	7649	
12-01-2000	Dow Jones	1134.45	1101	1176.42	1075.37	715127821	4.13	1.11	12.77	5.08	50	13344	305	1.19	108.59	96.15	1607.36	171.66	705	1.78	48	83.43	7071	
13-01-2000	S&P 500	1250.61	1271.27	1293.68	1216.82	90792126	8.14	2.12	13.28	2.3	116	20391	915	1.1	97.9	122.27	996.99	373.38	1138	6.33	1	53.43	8113	
14-01-2000	Dow Jones	4625.75	4629.28	4661.11	4577.74	595281205	8.26	9.76	8.8	9.41	73	529	165	1.4	111.37	45.89	2332.89	415.58	506	6.81	26	55.04	225	
15-01-2000	NASDAQ	1556.98	1570.51	1581.57	1544.3	866560662	-3.37	7.26	13.86	0.6	78	10163	4145	1.23	149.06	73.35	1063.49	386.73	1171	2.37	11	44.83	5201	
16-01-2000	S&P 500	3129.68	3099.33	3163.53	3055.58	415077918	1.43	9.15	12.98	4.42	64	18293	2151	0.98	120.13	88.91	1317.51	267.95	7930	3.2	15	39.56	10713	
17-01-2000	NASDAQ	2644.38	2615.55	2682.99	2569.35	965696140	1.97	3.04	11.51	9.3	53	28836	102	1.06	138.18	90.36	2143.77	487.03	4309	3.83	15	2.84	13591	
18-01-2000	NASDAQ	2389.37	2343.52	2413.03	2310	3991992	9.66	7.42	11.54	4.15	93	4179	424	0.91	98.43	113.38	2170.79	277.72	6698	6.03	2	62.21	14990	
19-01-2000	NASDAQ	4599.33	4581.54	4630.22	4534.98	215525643	5.17	7.14	3.37	1.38	114	15264	704	1.21	146.91	123.47	2332.18	492.52	4721	1.27	24	76.7	11054	
20-01-2000	NASDAQ	1087.29	1093.26	1116.08	1067.4	995760162	1.11	2.95	5.52	3.38	80	25604	4991	1.21	80.92	106.69	1645.57	467.25	6188	7.43	17	79.04	7061	
21-01-2000	S&P 500	3655.16	3690.95	3735.65	3607.51	992784081	-4.61	5.51	2.87	5.23	69	22031	2682	1.41	145.12	26.14	2477.43	262.86	4543	3.46	38	62.5	3790	
22-01-2000	S&P 500	4853.58	4870.27	4895.16	4818.44	349254868	3.48	6.27	2.11	2.26	59	625	2746	1.4	142.86	65.66	1677.52	164.4	1552	6.92	41	34.19	3259	
23-01-2000	NASDAQ	3240.67	3234.21	3282.76	3232.11	781089324	-3.56	4.81	4.06	4.93	60	5408	3222	1.39	84.71	94.42	2144.2	135.93	2553	9.33	39	2.36	6242	
24-01-2000	NASDAQ	4747.29	4792.6	4832.67	4701.95	695658158	7.52	8.27	13.45	8.45	77	28932	3214	1.18	122.34	86.02	1415.23	312.59	5599	8.7	31	89.59	3697	
25-01-2000	S&P 500	1209.03	1230.95	1251.48	1184.81	19640512	6.29	8.33	5.12	8.79	87	7142	3375	1.26	134.97	25.82	2400.2	405.28	5280	7.38	49	42.96	3414	
26-01-2000	S&P 500	2675.17	2718.19	2729.69	2652.15	834168124	1.91	8.51	14.61	0.97	73	20142	446	1.34	108.27	147.48	1964.53	170.42	6773	1.93	48	60.16	2677	
27-01-2000	S&P 500	2040.63	2043.39	2063.69	2012.31	97750245	1.44	8.03	9.76	2.03	88	9345	4254	0.86	86.43	144.29	931.07	227.28	3087	7.32	2	19.54	12464	
28-01-2000	Dow Jones	3923.28	3899.17	3964.59	3891.7	374017896	8.46	4.88	5.08	3.12	100	29309	2827	1.01	138.06	120.59	984.48	352.21	2889	6.36	7	14.28	4485	
29-01-2000	Dow Jones	4925.19	4880.47	4936.57	4857.01	67457001	-0.67	3.81	8.46	6.15	99	25025	2658	0.92	112.65	27.2	2314.38	284.35	300	7.73	4	8.77	11190	
30-01-2000	Dow Jones	2026.12	2048.73	2074.95	1884.56	447750702	-1.26	8.31	4.85	3.83	56	14148	3489	1.16	98.43	75.93	2176.25	341.55	7925	8.69	19	79.9	3724	

Figure 1: Screenshot of the raw (uncleaned) dataset



Date	Stock Inde	Open Pric	Close Pric	Daily High	Daily Low	Trading Volume	GDP Growth	Inflation Rat	Unemployme	Interest Ra	Consumer	Governme	Corporate	Forex USD	Forex USD	Crude Oil	Gold Price	Real Estat	Retail Sale	Bankruptc	Mergers &	Venture C	Consumer Spending (Billion
01-01-2000	Dow Jones	2128.75	2138.48	2143.7	2100.55	2670411	-0.37	6.06	6.1	6.06	114	27271	1645	1.04	119.87	47.2	1052.34	390.23	2229	2.12	3	76.64	4589
02-01-2000	S&P 500	2046.82	2036.18	2082.83	2009.53	690220415	3.19	4.95	6.62	2.19	101	16160	1008	1	98.22	52.84	1957.73	346.23	4156	1.4	21	5.67	10101
03-01-2000	Dow Jones	1987.92	1985.26	2022.28	1978.37	315284661	5.54	9.13	2.6	0.82	92	29962	4562	0.83	80.13	78.8	2339.49	439.46	340	0.79	48	39.43	13665
04-01-2000	Dow Jones	4625.02	4660.47	4665.26	4595.46	13098297	10	3.77	2.2	3.71	112	12745	4183	0.95	149.15	28.18	1308.54	213.07	8456	4.22	16	12.83	5192
05-01-2000	S&P 500	1998.18	1982.18	2044.31	1966.44	385306746	1.53	2.2	8.2	4.56	99	22293	3440	1.43	113.71	92.2	2210.08	405.49	1596	2.21	34	86.37	10688
06-01-2000	S&P 500	2087.8	2124.76	2153.18	2085.18	82664194	1.42	6.08	3.24	1.09	88	15880	2525	1.39	140	60.23	2338.39	441.06	403	0.82	45	96.8	8613
07-01-2000	Dow Jones	4037.59	3996.4	4055.78	3948.97	653722138	7.64	6.24	4.52	8.24	91	17830	3318	0.92	138.8	29.3	1309.15	319.6	4874	9.55	31	32.63	1303
08-01-2000	NASDAQ	2798.96	2826.64	2864.47	2766.89	423890033	7.08	4.39	13.01	3.77	66	20092	2265	1.06	136.86	117.92	896.46	265.51	1011	7.33	37	98.89	3771
09-01-2000	Dow Jones	4106.84	4141.59	4154.46	4062.99	867172199	5.97	4.87	9.42	9.18	92	27069	1113	1.02	112.13	104.95	2358.55	479.46	5193	5.18	30	75.26	13164
10-01-2000	Dow Jones	1261.46	1229.64	1296.14	1195.84	174582201	1.95	9.6	10.24	1.97	98	3940	3306	0.9	142.71	123.15	1361.82	167.59	6147	7.43	24	0.22	13774
11-01-2000	Dow Jones	2950.28	2943.31	2952.27	2925.46	320527008	-0.14	7.33	4.44	1.04	54	2045	1972	1.24	105.71	142.1	823.35	114.51	2861	6.11	9	48.58	7649
12-01-2000	Dow Jones	1134.45	1101	1176.42	1075.37	715127821	4.13	1.11	12.77	5.08	50	13344	305	1.19	108.59	96.15	1607.36	171.66	705	1.78	48	83.43	7071
13-01-2000	S&P 500	1250.61	1271.27	1293.68	1216.82	90792126	8.14	2.12	13.28	2.3	116	20391	915	1.1	97.9	122.27	996.99	373.38	1138	6.33	1	53.43	8113
14-01-2000	Dow Jones	4625.75	4629.28	4661.11	4577.74	595281205	8.26	9.76	8.8	9.41	73	529	165	1.4	111.37	45.89	2332.89	415.58	506	6.81	26	55.04	225
15-01-2000	NASDAQ	1556.98	1570.51	1581.57	1544.3	866560662	-3.37	7.26	13.86	0.6	78	10163	4145	1.23	149.06	73.35	1063.49	386.73	1171	2.37	11	44.83	5201
16-01-2000	S&P 500	3129.68	3099.33	3163.53	3055.58	415077918	1.43	9.15	12.98	4.42	64	18293	2151	0.98	120.13	88.91	1317.51	267.95	7930	3.2	15	39.56	10713
17-01-2000	NASDAQ	2644.38	2615.55	2682.99	2569.35	965696140	1.97	3.04	11.51	9.3	53	28836	102	1.06	138.18	90.36	2143.77	487.03	4309	3.83	15	2.84	13591
18-01-2000	NASDAQ	2389.37	2343.52	2413.03	2310	3991992	9.66	7.42	11.54	4.15	93	4179	424	0.91	98.43	113.38	2170.79	277.72	6698	6.03	2	62.21	14990
19-01-2000	NASDAQ	4599.33	4581.54	4630.22	4534.98	215525643	5.17	7.14	3.37	1.38	114	15264	704	1.21	146.91	123.47	2332.18	492.52	4721	1.27	24	76.7	11054
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25-01-2000	S&P 500	1209.03	1230.95	1251.48	1184.81	19640512	6.29	8.33	5.12	8.79	87	7142	3375	1.26	134.97	25.82	2400.2	405.28	5280	7.38	49	42.96	3414
26-01-2000	S&P 500	2675.17	2718.19	2729.69	2652.15	834168124	1.91	8.51	14.61	0.97	73	20142	446	1.34	108.27	147.48	1964.53	170.42	6773	1.93	48	60.16	2677
27-01-2000	S&P 500	2040.63	2043.39	2063.69	2012.31	97750245	1.44	8.03	9.76	2.03	88	9345	4254	0.86	86.43	144.29	931.07	227.28	3087	7.32	2	19.54	12464
28-01-2000	Dow Jones	3923.28	3899.17	3964.59	3891.7	374017896	8.46	4.88	5.08	3.12	100	29309	2827	1.01	138.06	120.59	984.48	352.21	2889	6.36	7	14.28	4485
29-01-2000	Dow Jones	4925.19	4880.47	4936.57	4857.01	67457001	-0.67	3.81	8.46	6.15	99	25025	2658	0.92	112.65	27.2	2314.38	284.35	300	7.73	4	8.77	11190
30-01-2000	Dow Jones	2026.12	2048.73	2074.95	1984.56	447750702	-1.26	8.31	4.85	3.93	56	14148	3489	1.16	98.43	75.93	2176.25	341.55	7925	8.69	19	79.9	3724

Figure 2: Screenshot of the cleaned dataset

4.2 Visualization / Output Results

Presented the key visualizations, dashboards, or generated content produced by your project.

1.Maximum of GDP Growth

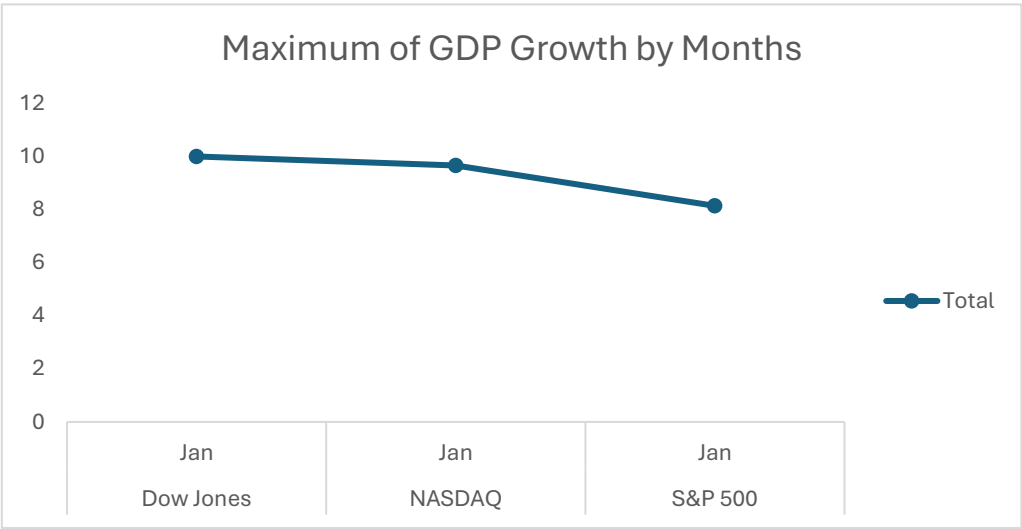

Figure 3: Maximum of GDP Growth

2.Average of Trading volume by quarters

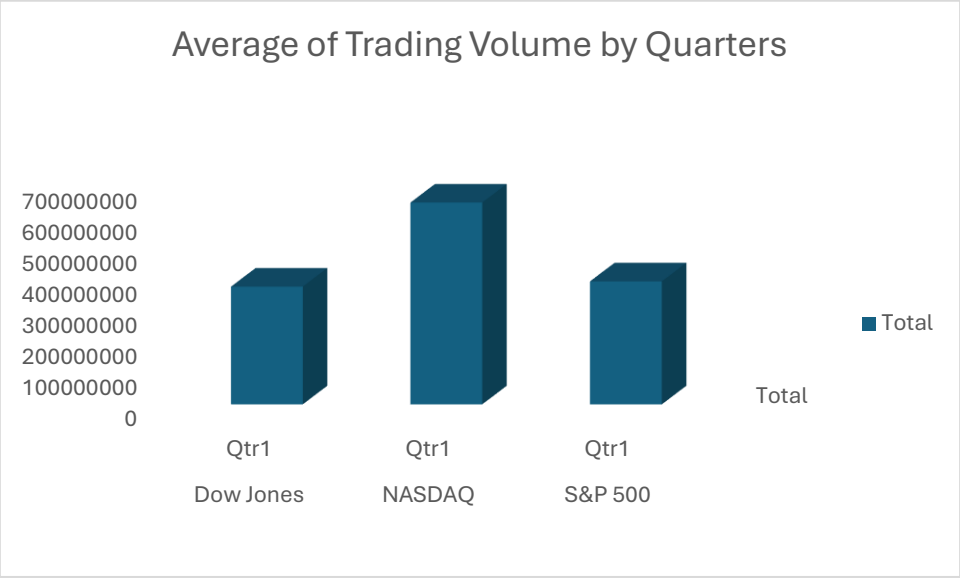

Figure 4: Average of Trading Volume By quarters

3.Maximum of unemployment rate by years

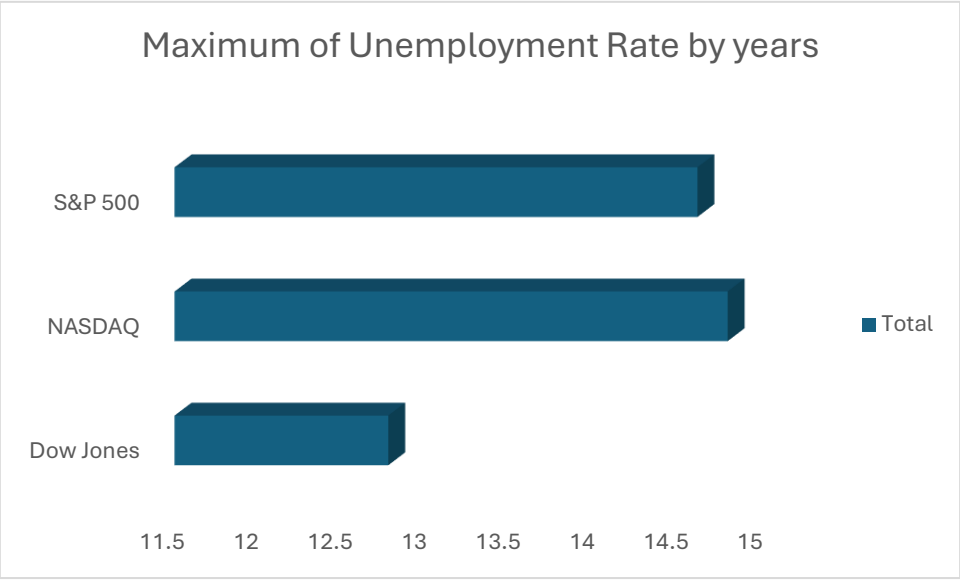

Figure 5: Maximum of Unemployment rate by years

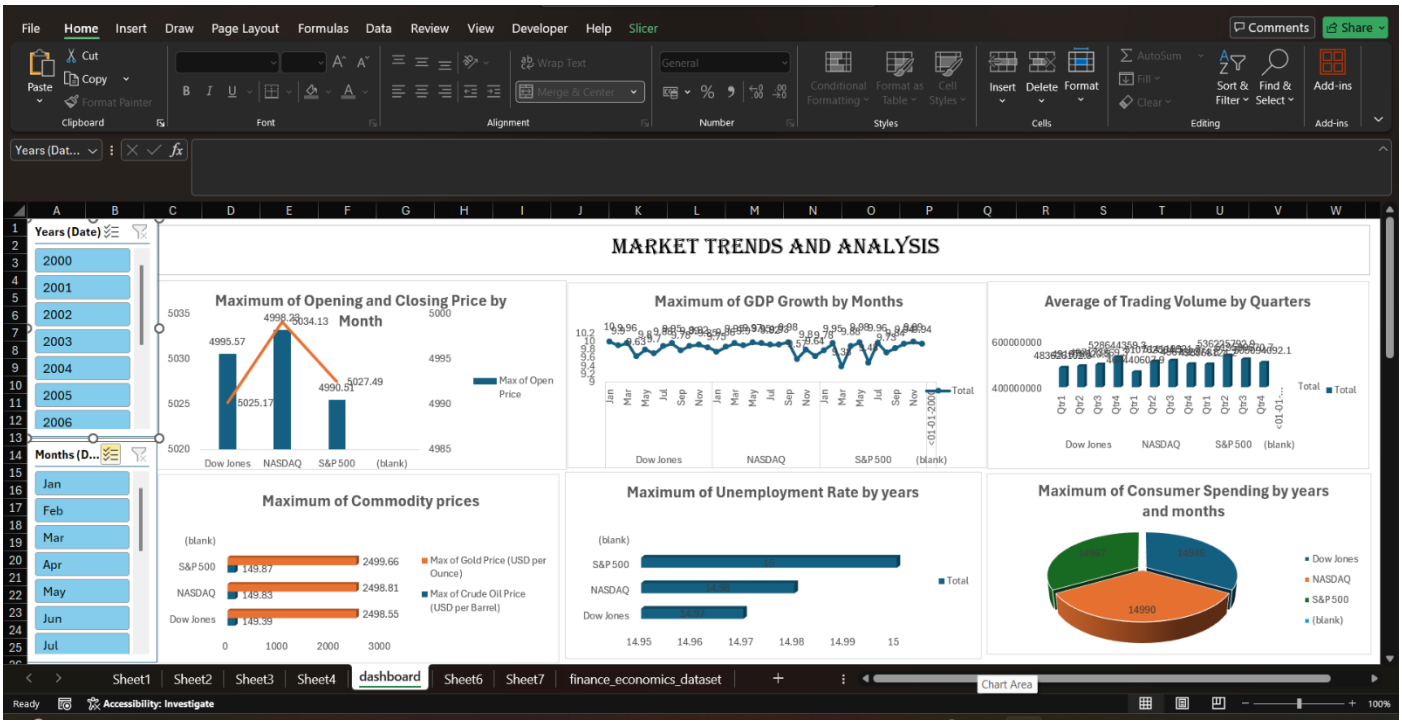

Figure 6: screenshot of Main Dashboard

4.3 AI Tool Integration Results

Artificial Intelligence tools were integrated into the project to enhance data analysis, automate tasks, and improve the overall quality of insights and reporting.

Tool used:
*The primary AI tool used in this project was **ChatGPT**, along with support from **Microsoft Copilot** for data-related suggestions and analysis.*

Prompts used:
The following key prompts were used during the project:

- *"Analyze this finance dataset and identify key trends in GDP growth, inflation, and stock market performance."*
- *"Suggest data cleaning steps for a dataset with financial and economic indicators."*
- *"Generate insights and summary report based on the cleaned dataset."*

Output quality:
*The AI-generated outputs were **highly relevant, structured, and easy to understand**. The tool provided clear explanations, meaningful insights, and well-organized content. It also helped in identifying patterns and suggesting improvements in data cleaning and visualization.*

How it enhanced the project:
The use of AI tools significantly improved the efficiency and effectiveness of the project in the following ways:

- *Reduced the **time required for data analysis and report writing***
- *Provided **quick and accurate suggestions** for data cleaning and processing*
- *Helped generate **professional-quality content** for the report*
- *Assisted in **identifying trends and insights** that might be difficult to detect manually*
- *Improved overall **presentation and clarity** of the project*

  The integration of AI tools demonstrates how modern technologies can be effectively used to enhance data analytics and decision-making processes.

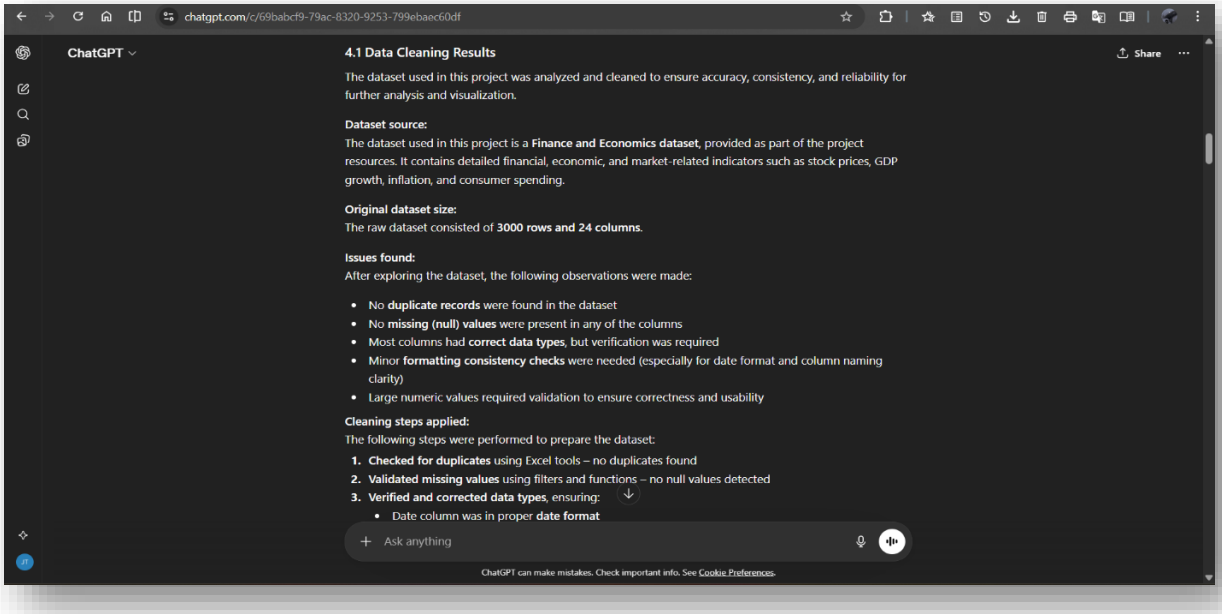

Figure:7 Screenshot of AI Tool using during the Project

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The project generated several meaningful insights after data cleaning, analysis, and visualization:

- *The dataset was found to be **clean and well-structured**, with no missing values or duplicate records, ensuring high data accuracy.*
- *The **GDP growth trend** showed clear patterns of economic expansion and slowdown over time.*
- *The **inflation analysis** helped identify periods of rising prices, which can impact purchasing power and economic stability.*
- *The **stock market trend visualization** revealed fluctuations and volatility, reflecting changes in market conditions and investor behavior.*
- *The **relationship between interest rates and investment** indicated how financial decisions are influenced by economic factors.*
- *AI tools significantly reduced the **time required for analysis and report preparation**, improving efficiency and productivity.*

5.2 Project Links

1.finance_economics_dataset (1).ods

5.3 Limitations

Despite successful implementation, the project has the following limitations:

- *The **dataset used is static and does not include real-time updates.***
- *The analysis is limited to **descriptive analysis** and does not include predictive or advanced statistical modelling.*
- *Some economic relationships may require **deeper domain knowledge** for accurate interpretation.*
- ***Limited access to advanced AI features restricted the full potential of automation.***

5.4 Future Work

The project can be further enhanced in the following ways:

- *Integrate **real-time data sources** using APIs or automated tools*
- *Apply **machine learning models** for forecasting and trend prediction*
- *Develop a more **interactive and dynamic dashboard***
- *Expand the dataset with more **global financial and economic indicators***
- *Create a **standardized AI prompt library** for faster and better analysis*

5.5 Conclusion

This project successfully demonstrated how financial and economic data can be effectively cleaned, analyze, and visualized using tools like Microsoft Excel, Power BI, and AI technologies. *The main objective of the project—to transform raw data into meaningful insights—was successfully achieved.*

The use of AI tools played a significant role in improving efficiency, reducing manual effort, and enhancing the quality of outputs. *The visualizations created in this project provide clear insights into economic trends, helping in better understanding and decision-making.*

*Overall, this project helped in developing practical skills in data analytics, visualization, and AI integration**. It highlights the importance of combining traditional tools with modern AI technologies in solving real-world problems**.*

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

Title of Paper	Author(s)	Year	Download / Access Link
Stock Returns, Expected Returns, and Real Activity	Eugene F. Fama	1990	https://didattica.unibocconi.it/mypage/upload/FAMA20081107164805.PDF
The Time-Series Relations Among Expected Return, Risk, and Book-to-Market	Eugene F. Fama and Kenneth R. French	1999	https://www.sciencedirect.com/science/article/pii/S0304405X99000306